Java Overview

Java is a high-level, object-oriented, platform-independent programming language. Features: Simple, Secure, Robust, Portable, Multithreaded, Distributed, and High Performance. JVM executes Java bytecode, enabling Write Once, Run Anywhere (WORA).

Variables and Data Types

Primitive Types: byte, short, int, long, float, double, char, boolean. Reference Types: Arrays, Classes, Interfaces, Strings. Variables: Local, Instance, Static. Type casting can be implicit or explicit.

Operators and Control Statements

Operators: Arithmetic, Relational, Logical, Assignment, Bitwise, Ternary. Control Statements: - if, if-else, switch - for, while, do-while - break and continue

Object-Oriented Programming

OOP Concepts: 1. Class and Object 2. Encapsulation 3. Inheritance 4. Polymorphism 5. Abstraction These concepts improve code reusability and maintainability.

Arrays and Strings

Arrays store multiple values of the same type. One-dimensional and multidimensional arrays are supported. String is immutable. Common String methods: length(), substring(), equals(), toUpperCase().

Exception Handling

Keywords: try, catch, finally, throw, throws. Types: - Checked Exceptions - Unchecked Exceptions
Exception handling prevents abrupt program termination.

Collections Framework

Important Interfaces: - List - Set - Queue - Map Common Classes: ArrayList, LinkedList, HashSet, TreeSet, HashMap.

Multithreading and File Handling

Multithreading allows concurrent execution of tasks. Thread creation: Thread class or Runnable interface. File Handling: Read/write files using File, FileReader, BufferedReader, FileWriter.